CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. When ice at 0 °C is heated, the temperature stays at 0 °C till all ice melts. This shows that: 1

(a) the thermometer is faulty **(b)** ice cannot be heated
(c) the heater is weak **(d)** heat is used to change the state

2. Water of 100 g at 0 °C is converted to ice at 0 °C. The mass of ice formed is: 1

(a) less than 100 g **(b)** 0 **(c)** 100 g **(d)** more than 100 g

3. A student claims that the water in a cup disappeared. The correct statement is: 1

(a) it was destroyed **(b)** it became oxygen
(c) it vanished **(d)** it changed to vapour and mixed with air

4. A student heats 100 g of ice and records temperatures. The graph has a flat portion at 0 °C and another at 100 °C. The flat portions represent: 1

(a) heating of solid and liquid **(b)** freezing
(c) cooling **(d)** melting and boiling

5. If 50 mL of water evaporates from a vessel in 5 days, the average loss per day is: 1

(a) 10 mL **(b)** 50 mL **(c)** 250 mL **(d)** 5 mL

6. Why does a puddle disappear faster than a lake in the same sun? 1

(a) smaller volume and large surface to depth ratio
(b) lake has no water
(c) lake is salty
(d) puddle is hotter

7. Which factor does not affect the rate of evaporation? 1

(a) wind **(b)** temperature
(c) the colour of the container **(d)** area exposed

8. A tank of 1000 L loses 2 L per day by evaporation. After 30 days, the water lost is: 1

(a) 2 L **(b)** 600 L **(c)** 60 L **(d)** 30 L

9. **Assertion (A):** Evaporation takes place at all temperatures. 1
Reason (R): Boiling takes place only at a fixed temperature, about 100 degrees C.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Wet clothes dry faster on a windy day. 1
Reason (R): Moving air carries away water vapour quickly, so more water can evaporate.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why a wet cloth kept in a closed polythene bag does not dry. 2

12. How does a glass of cold drink 'sweat' even though the glass has no holes? 2

13. How would water table levels change with excessive pumping? 2

14. Why does water in a pressure cooker boil above 100 °C? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A tank of 2000 L loses 4 L per day by evaporation. If it is filled once a month (30 days) how much is lost, and what percentage? 3

16. Why do hill stations often receive more rainfall? 3

17. Describe two ways in which forests help the water cycle. 3

OR

Explain with an experiment how to prove that the water which disappears from a pond goes to the air. 3

18. Describe how an igloo can survive in cold weather and not melt. 3

OR

Explain why a layer of oil on a water tank reduces evaporation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Nandini in Kochi spreads her wet salwar on a rope in the sun and puts a similar wet cloth rolled up in a shaded corner. After two hours the cloth in the sun is dry, while the other is still wet. On a rainy day, even the cloth in the sun takes much longer to dry. 5
(i) Name the process by which the wet cloth dries.
(ii) Name one factor that helped the cloth in the sun dry sooner.
(iii) What happens to the water that leaves the cloth?
(iv) Why does the cloth dry slowly on a rainy day?

20. Explain how a solar still can convert sea water to drinking water, naming the changes of state. 5

OR

Discuss why water conservation is both a personal and a national responsibility with five examples. 5

END OF QUESTION PAPER — Total: 40 Marks